Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Expand ($x-3)^2$.
2. Factorise x^2-16.
3. Solve ($x-2)(x+5)=0$.
4. Solve x^2 = 9 in $\mathbb{R}$.
5. Give the sign of 2 $x-6$.
6. Determine the minimum of ($x-1)^2+4$.

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Present two methods for factorising a quadratic. In the proof by discriminant, name the three cases and explain why they are exhaustive.

Investigate before formalising

A rectangle has an area of 24 m ^{2} and a perimeter of 20 m. Without randomly trying all the dimensions, determine its sides. Start with a drawing and explain the two constraints.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.